SCRUM 4

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Project Goals

- Use Socket.io to create a real time voting app
- Broadcast data to all connected client servers
- Update webpage dynamically
- Allow clients to answer a live poll while displaying immediate results

Methods Used

- Server created through index.js
  - Node.js on the backend to manage votes and send updates via Socket.io
- App created using index.html and app.js
  - Connects the browser to the server using Socket.io
  - Object.values().reduce() - calculates percentages in the progress bars

Team Roles

- Yacob: Server-side development (index.js)
- Victoria: Front-end development (index.html, app.js)
- Amanda(Scrum Master): Testing and presentation slides

Code Illustration:

Server Setup

```
const http = require('http');
const server = http.createServer();
const io = require('socket.io')(server, { cors: { origin: '*' } });
```

Vote storage

```
let votes = { JavaScript: 0, Python: 0, Java: 0, 'C#': 0 };
```

Handling new connections

```
io.on('connection', (socket) => {
    console.log('a user connected');
    socket.emit('updateVotes', votes);
```

Handling votes

```
socket.on('vote', (option) => {
    if (votes[option] !== undefined) {
        votes[option]++;
        io.emit('updateVotes', votes);
    }
});
```

Starting the server

```
server.listen(8080, () => {
    console.log('listening on http://localhost:8080');
});
```

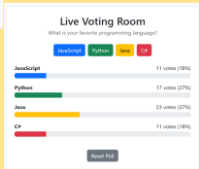
```
const socket = io('ws://localhost:8080');
```

```
socket.on('updateVotes', (votes) => {
    const total = Object.values(votes).reduce((a, b) => a + b, 0);
    const resultsDiv = document.getElementById('results');
    resultsDiv.innerHTML = '';

    Object.entries(votes).forEach(([option, count]) => {
        const count = votes[option] || 0;
        const percent = total > 0 ? Math.round((count / total) * 100) : 0;
        const color = colors[index];

        const item = document.createElement('div');
        item.className = 'mb-2';
        item.innerHTML = `
            <div class="d-flex justify-content-between mb-1">
                <span><strong>${option}:</strong></span>
                <span>${count} vote${count !== 1 ? 's' : ''} (${percent}%)</span>
            </div>
            <div class="progress">
                <div class="progress-bar bg-${color}" style="width: ${percent}%; transition: width 0.6s ease;"></div>
            </div>
        `;
        resultsDiv.appendChild(item);
    });
});
```


Testing



- Functionality testing:
  - Opening multiple browsers
  - Vote count accuracy
  - Resetting vote count
- Connection testing:
  - Terminal should show successful connection message
  - Is the server running?